What is Velocity?

A Journey Through Physics

Introduction

Let's define velocity!

Introduction (cont.)

$$
\begin{aligned}
V \equiv (Q_\xi, X) &= \mathfrak{X}_{Q_\xi}[X] \\
\delta[Q_\xi] = K(\delta, \mathfrak{X}_{Q_\xi}) &= \Omega_{\text{ext}}(\delta, \mathfrak{X}_{Q_\xi}) - G(\delta, \mathfrak{X}_{Q_\xi}) \\
&= \Omega(\delta, \mathfrak{L}_\xi) + Q_{[\mathbb{X}(\delta), \mathbb{X}(\mathfrak{L}_\xi)]} + \oint_{\partial\Sigma} \mathbb{X}(\mathfrak{L}_\xi) \lrcorner\, \vartheta(\delta) \\
&\quad + \oint_{\partial\Sigma} \mathbb{X}(\mathfrak{L}_\xi) \lrcorner\, \vartheta(\delta) - G(\delta, \mathfrak{L}_\xi) \\
&= \Omega_{\text{ext}}(\delta, \mathfrak{L}_\xi) - G(\delta, \mathfrak{L}_\xi)
\end{aligned}
$$

Introduction (cont.)

$$
\begin{aligned}
&= \delta\left[\int_\Sigma \left(\vartheta(\mathfrak{L}_\xi) - \xi \lrcorner L\right)\right] - Q_{\delta[\xi]} - \oint_{\partial\Sigma} \xi \lrcorner \vartheta(\delta) \\
&\quad + Q_{[\mathbb{X}(\delta),\mathbb{X}(\mathfrak{L}_\xi)]} + \oint_{\partial\Sigma} \mathbb{X}(\mathfrak{L}_\xi) \lrcorner \vartheta(\delta) \\
&= \delta\left[\int_\Sigma \left(\vartheta(2\nabla_{(a}\xi_{b)}, \mathcal{L}_\xi \psi^I) - \xi \lrcorner L\right)\right] - Q_{\delta[\xi]} \\
&\quad + Q_{[\mathbb{X}(\delta),\xi]} + \oint_{\partial\Sigma} \xi \lrcorner \vartheta(\delta)
\end{aligned}
$$

Introduction (cont.)

$$\dot{F} = (Q_\xi, F)$$

$$(Q_\xi, g_{ab}) = \mathcal{L}_\xi g_{ab} = 2\nabla_{(a}\xi_{b)}$$

$$(Q_\xi, \psi^I) = \mathcal{L}_\xi \psi^I$$

$$(Q_\xi, x^\mu) = \xi^\mu$$

$$\Gamma^c{}_{ab} = \tfrac{1}{2} g^{cd}(\partial_a g_{db} + \partial_b g_{da} - \partial_d g_{ab})$$

$$g^{ac} g_{cb} = \delta^a{}_b, \qquad \nabla_c g_{ab} = 0$$

$$(Q_\xi, Q_\xi) = -G(\mathfrak{L}_\xi, \mathfrak{L}_\xi) = \oint_{\partial\Sigma} \xi \lrcorner\, \vartheta(\mathfrak{L}_\xi)$$

. . . uhm, well. . . maybe I am wrong about that.

Basic Definition

Velocity can generally be defined as the change in distance traveled divided by the time it takes to travel:

$$v = \frac{\Delta x}{\Delta t}$$

Taking the infinitesimal limit:

$$v = \frac{\mathrm{d}x}{\mathrm{d}t}$$

Velocity also contains direction. The easiest way to represent it is as a vector:

$$\dot{\mathbf{r}} = \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t} = \left\langle \frac{\mathrm{d}x}{\mathrm{d}t}, \frac{\mathrm{d}y}{\mathrm{d}t}, \frac{\mathrm{d}z}{\mathrm{d}t} \right\rangle$$

Newtonian Mechanics

Newton's second law:

$$\mathbf{F} = m\frac{\mathrm{d}^2\mathbf{r}}{\mathrm{d}t^2}$$

Solving for velocity:

$$\dot{r} = \int \frac{\mathbf{F}}{m}\, dt$$

Keeping track of the reference frame is simple:

$$\dot{r}' = \int \frac{\mathbf{F}}{m}\, dt - \dot{r}_{\text{reference frame}}$$

Newtonian Mechanics (cont.)

Relying on frames of reference creates problems (e.g. fictitious centrifugal forces).
One way to make Newtonian mechanics generally covariant is the **Newton–Cartan** approach:

- Space-time structure similar to general relativity
- But time remains absolute

Velocity is then defined as

$$V^\mu = \frac{\mathrm{d}x^\mu}{\mathrm{d}t} = (1,\ \frac{\mathrm{d}x^i}{\mathrm{d}t})$$

While this looks simple, it is an extreme simplification.

Special Relativity

In special relativity, time becomes another relative dimension.
The velocity vector becomes four-dimensional:

$$U^\mu = \frac{\mathrm{d}x^\mu}{\mathrm{d}\tau}$$

where τ is **proper time** (the time experienced by the traveler).
Galilean relative velocity

$$v_{\mathrm{rel}} = v_1 - v_0$$

gains relativistic corrections:

$$v_{\mathrm{rel}} = \frac{v_1 - v_0}{1 - v_1 v_0 / c^2}$$

General Relativity

Absolute acceleration creates paradoxes and preferential frames.
Combining the covariant Newtonian framework with Einsteinian proper time yields the same-looking equation

$$U^\mu = \frac{\mathrm{d}x^\mu}{\mathrm{d}\tau}$$

but now with the normalization

$$g_{\mu\nu} U^\mu U^\nu = -c^2.$$

The geometry of spacetime is curved, further separating observers. The rest proceeds as in special relativity, but with additional corrections expressible in the language of Riemannian geometry and geodesics.

Warp Drives

Warp drives *move* faster-than-light motion.

- Most solutions have the matter itself not moving at all.
- Spacetime contracts in front and expands behind the ship.
- Proper velocity of the ship is essentially 0, yet an external observer sees it traverse great distances in short coordinate time.

This makes the definition of "velocity" rather challenging to define.

Dynamics — Phase Space

In phase space the fundamental coordinates are generalized positions q^i and momenta p_i. If the flow is generated by a Hamiltonian H, then

$$v = \dot{q}^i = \frac{\partial H}{\partial p_i}.$$

Velocity is simply one component of the Hamiltonian vector field on phase space.

Poisson Structure

We introduce the Poisson bracket

$$\dot{f} = \{f, H\} = \frac{\partial f}{\partial q^i}\frac{\partial H}{\partial p_i} - \frac{\partial f}{\partial p_i}\frac{\partial H}{\partial q^i}.$$

The entire vector field can then be written

$$X_H = \{\,\cdot\,, H\}.$$

The underlying geometry is symplectic (we will not go into the details here).

Other Approaches (Speed Run)

- **Fluid dynamics**: a vector field that assigns a Newtonian velocity to every point in space.
- **Quantum mechanics**: velocity becomes a probability current

$$\mathbf{v} = \frac{\mathbf{j}}{|\psi|^2}.$$

- **Kinetic theory**: combines the probabilistic nature of QM with the continuum of fluids.
- **Geometric mechanics**: velocity is a "natural element of a tangent bundle."

Thank you